Lab work MA21
Fluorescence and quantum efficiency

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Work expected:

During this lab work, you will be asked to actively participate by answering question orally and in the spread-sheet. Morever, there will be questions to prepare beforehand. You will be asked to hand out a detailed report which will be evaluated according to the quality of the answers to the questions and the clarity of explanations of the experiments. Each report must be handed out at the end of the lab work.

Overview of the lab work: some materials can emit light upon external excitation in particular by light. This lab work will consist in studying the so-called fluorescence of some materials using spectroscopy and to evaluate the so-called quantum efficiency of a dye molecule and of a light emitting diode (LED) using an integrating sphere.

This lab work will be done in 4h and a report will be handed out at the end of the 4h.

Lab equipement :

-Integrating sphere

-Spectrometer Ocean Optics + fibre + PC

-Power supply

-Photodiode for integrating sphere

-Rhodamine solution

-Green laser

- LED (blue)

-White light source: incandescent and white LED

1st part: Fluorescence and spectroscopy:

Some materials can emit light upon external excitation which can be by optical means. In this case we talk about fluorescence or luminescence. The basic principle is that a material absorbs some external energy coming from external photons which will promote an electron from a lowest energy band (ground state) to a higher one (excited state). When the electron comes back to the lowest energy, it can do so by releasing its energy through the emission of a photon. This is fluorescence. This is done at the atomic level and can be quite complex in molecules. Figure 1 presents a schematic of the fluorescence of a molecule.

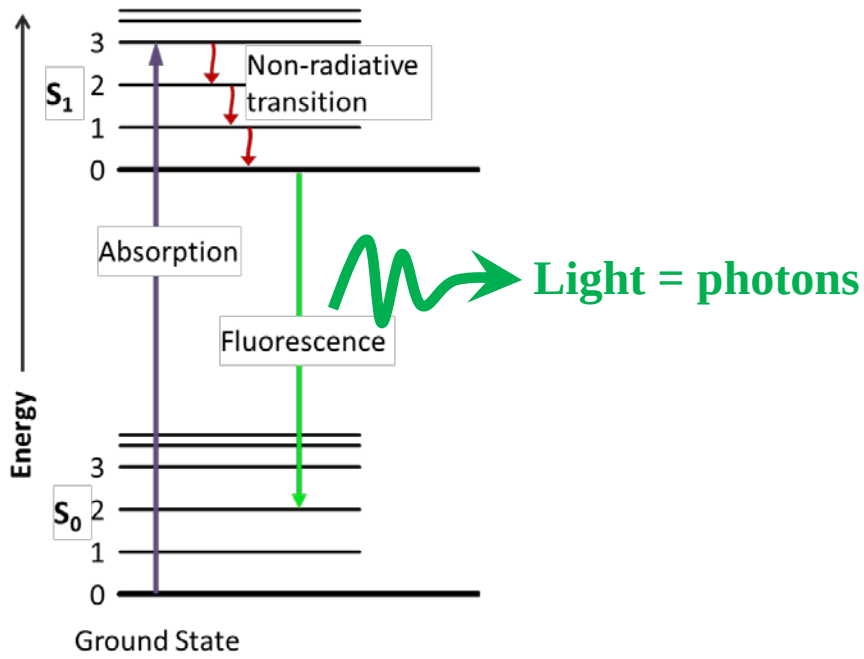

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Figure 1: Jablonski diagram of a molecule

Figure 1 is a so-called Jablonsky diagram of a molecule which can undergo via absorption of light a change of energy level from S_0 to S_1 and then back again to S_0 with the emission of a photon. In this case, the following equations hold:

Light absorption: $$S_0 + h\nu_{ex} \rightarrow S_1$$

Light emission: $$S_1 \rightarrow S_0 + h\nu_{em} + heat_2$$

Where S_0 is the degenerate ground state level of the molecules, S_1 is the degenerate excited state level of the molecules and u_{ex} and u_{em} are the excitation and emission frequencies, respectively.

Experimental work:

We will use an Ocean Optics spectrometer with an optical fiber as a probe. The software OceanView will be used.

1- Start OceanView and get used to the software. Look for the following settings:
   -Continuous run
   -One shot run mode
   -View data in a table that you can then export into excel for the quantum efficiency measurement. You will need the excel file for the quantum efficiency to calculate the intensity area under the spectra.

-Use the ‘print screen’ option to obtain your spectra or export the data table into excel to make a graph

See Figure 1 for an overview of the spectrometer. We note that the light gets inside the spectrometer using the blue optical fiber. You will need to orientate it in order to get the different light sources inside the spectrometer.

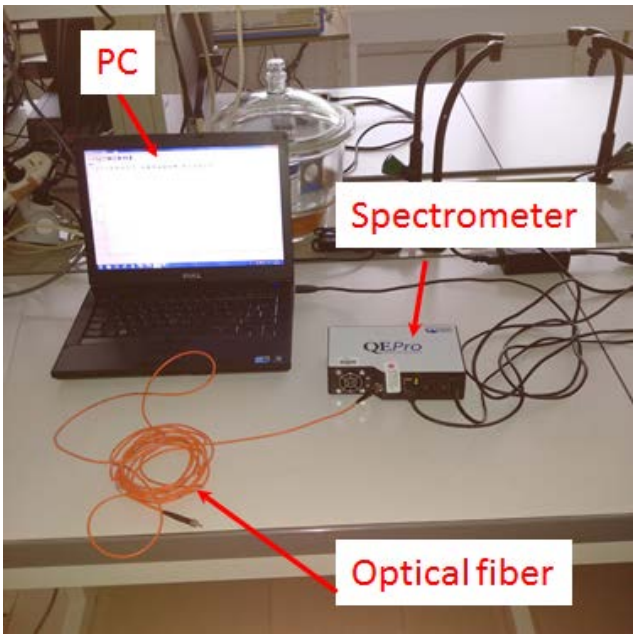

Figure 1: Picture of the connections for the spectrometer.

2- Take various spectra: room lights, your mobile light, a light bulb provided, LEDs provided (See Figure 2 for the electrical connections) and the laser provided (see Figure 3). You will look at the range 350-800 nm for the measurements. Adjust the intensity level on the graphs in order to have nice and observable spectra.

/!\ The laser emission is dangerous. Make sure the optical density is glued on the laser before switching it on. It will be removed later on for the quantum efficiency measurements.

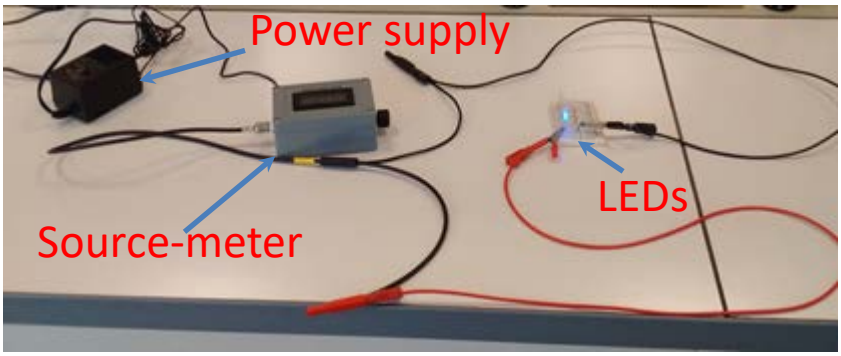

Figure 2: Picture of the connections for the LEDs.

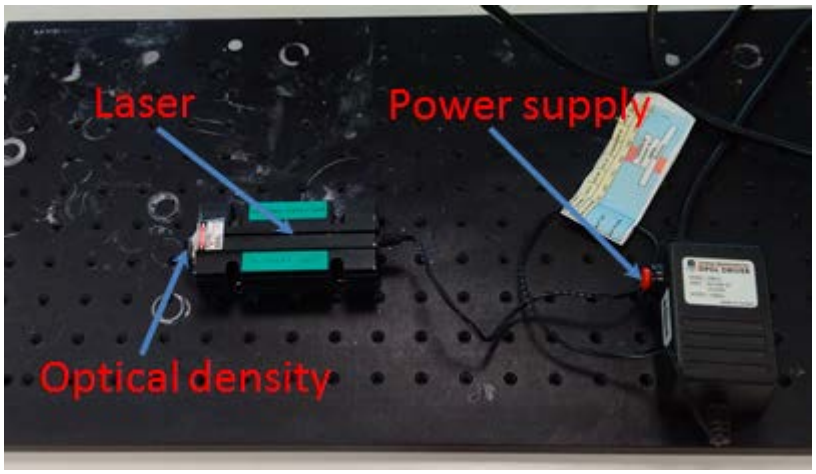

Figure 3: Picture of the laser configuration

3- What can you say about these different sources of light? What makes them different from each other, what physical effects occur?

4- What is the fundamental relation between $h\nu_{ex}$ and $h\nu_{em.}$

5- Take the absorption measurement of the Rhodamine solution with the spectrofluorometer (Hach spectrofluorometer, see Lab work/TP on Transmission to know how to operate it).

6- Take the spectrum of Rhodamine using the green laser provided as the excitation source. See Figure 4 for the general layout of the measurement to be done. What is the best configuration between light excitation and light collection? In some configuration, you seem to find a narrow peak at 535 nm or so. What is it? Why do you have it?

7- Compare your findings of absorption and emission of Rhodamine with the literature.

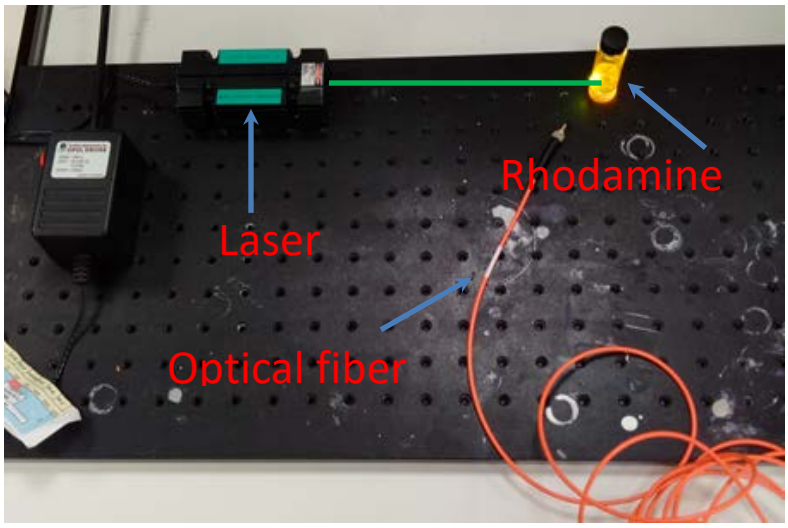

Figure 4: Picture of the rhodamine excitation by the laser.

2nd part: Quantum efficiency:

The fluorescence quantum efficiency (QE) gives the efficiency of the fluorescence process. It is defined as the ratio of the number of photons emitted to the number of photons absorbed by a given molecule/dye/absorber. It is given by:

$$\Phi = \frac{\text{Number of photons emitted}}{\text{Number of photons absorbed}}$$

The maximum fluorescence quantum efficiency is 1.0 (100%) as it means each photon absorbed results in a photon emitted. QEs of 0.10 are still considered quite fluorescent. Another way to define the quantum efficiency of fluorescence, is by the rate of excited state decay and is given by

$$\Phi = \frac{k_f}{\sum_i k_i}$$

where k_f is the fluorescent decay rate and k_i are all the possible decay rates, including k_f and all the non-radiative ones (see figure 1 previously). We will measure the Qe of two different samples, a solution of rhodamine (dye molecule in ethanol) and a blue LED, electrically driven. In order to measure both QE, we will use an integrating sphere which can collect all the light in all directions emitted by a given emitter. We will describe the case of the rhodamine optically excited using a 532 nm green laser.

Principle of an integrating sphere

Originally developed by J. C. de Mello *et. al.* [ref], the method of QE measurement involves the use of an integrating sphere, which is a hollow sphere with its inner surface coated by a highly scattering material. When a light source is placed inside, the light is isotropically redistributed over the sphere interior-surface regardless of the angular dependence of the emission. The basic experimental set up is shown in Figure 2: an output optical fibre is plugged onto an open port of the sphere. In front of the optical fibre is a baffle to prevent possible direct illumination from the light source. Another light source, e.g. excitation laser, can be directed into the sphere through a small entrance hole. The fibre will be plugged to the previously used spectrometer.

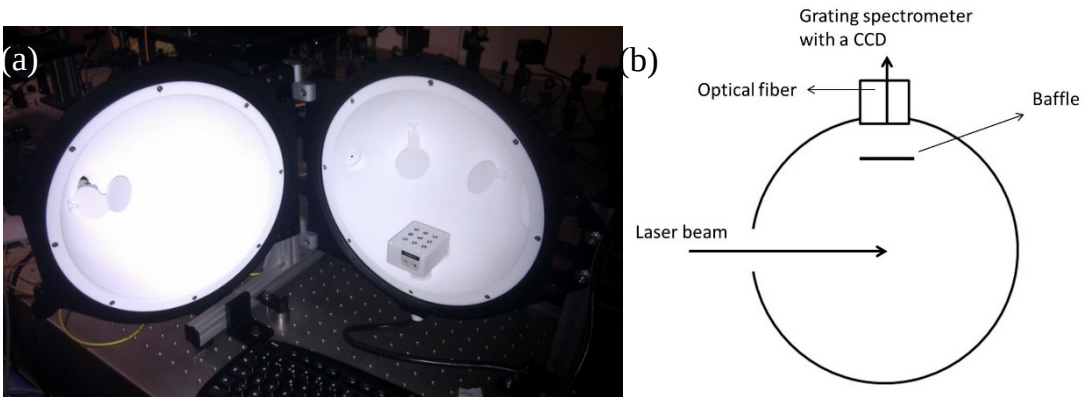

Figure 2 Basic experimental setup for EQE measurement: (a) An integrating sphere; (b) The schematics of the measurement system.

The measurement is determined by a three-step procedure, as shown in Figure 3. The total amount of laser light striking the sample can be separated into two contributions: the incident laser beam which strikes the sample directly; diffused laser light which is scattered from the wall of the sphere and subsequently strikes the sample over its entire surface.

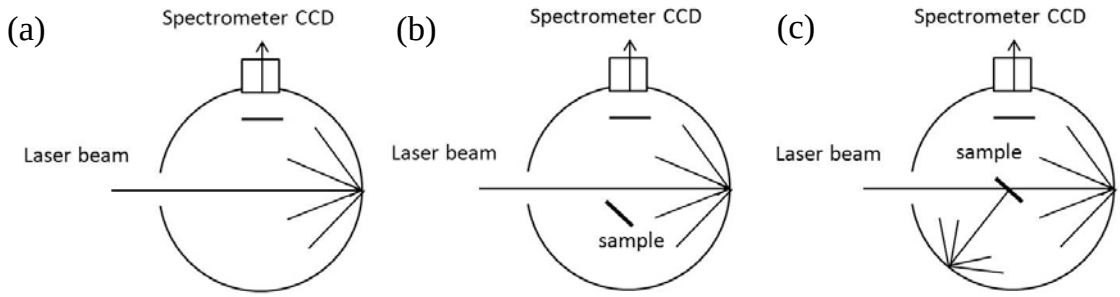

Figure 3 Three-step measurement procedure

In the second step (Figure 3-b), we assume that a fraction *μ* of the laser light scattered from the sphere wall is absorbed by the sample. In the third step (Figure 3-c), the laser beam first strikes the sample. A fraction, *A*, of this incident light will be absorbed by the sample. A fraction, *1-A*, will either be transmitted or reflected. The unabsorbed light strikes the interior surface of the sphere from which it is scattered. A fraction, *μ*, of this scattered light is then absorbed by the sample. We can therefore write:

$$S_2^{\text{Laser}} = (1-\mu) S_1^{\text{Laser}}$$

$$S_3^{Laser} = (1-A)S_2^{Laser} = (1-A)(1-\mu)S_1^{Laser}$$

where S_1^{Laser}, S_2^{Laser} and S_3^{Laser} are the intensities of laser in steps 1, 2, and 3 of Figure 3 respectively, as the example given in Figure A-3.

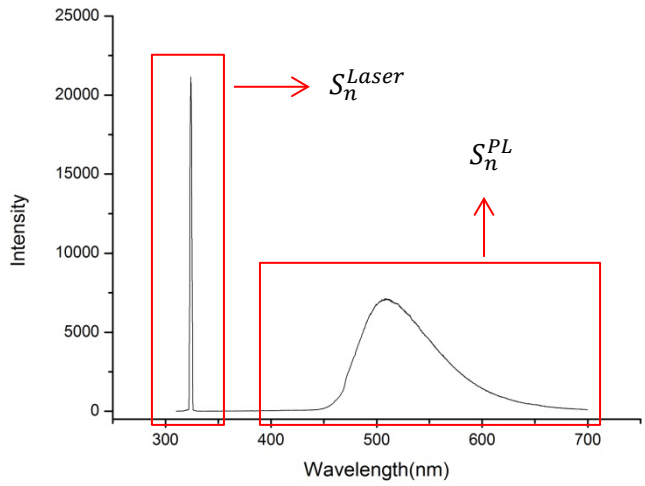

Figure 4: Profiles of S_n^{Laser} and S_n^{PL} in a measured spectrum (this is an example and different from the rhodamine case)

From these equations, we obtain an expression for the absorption coefficient:

$$A = \left(1 - \frac{S_3^{\text{Laser}}}{S_2^{\text{Laser}}}\right)$$

In the second step, the detected signal consists of a scattered laser light and the PL of the sample due to the absorption of the scattered laser light. The number of photons which strike the detector is given by $S_2^{\text{Laser}} + S_2^{\text{PL}}$.

In the third step, a fraction of the incident laser light is reflected by the sample, a fraction is transmitted, and the remaining fraction, *A*, is absorbed. Assuming that the location from which the laser light is first scattered is unimportant, the contribution of scattered laser light to the measured spectrum is given by:

$$(1 - A)\left(S_2^{\text{Laser}} + S_2^{\text{PL}}\right)$$

The PL by the absorption of the collimated laser light ($QE.A.S_1^{Laser}$) also contributes to the emission profile measured in the third step. Hence the total intensity over the spectrum in the third step is given by:

$$S_3^{Laser} + S_3^{PL} = (1 - A)(S_2^{Laser} + S_2^{PL}) + QE.A.S_1^{Laser}$$

Rearranging this equation and eliminating S_2^{Laser} and S_3^{Laser}, we find:

$$QE = \frac{S_3^{PL} - (1 - A)S_2^{PL}}{A.S_1^{Laser}}$$

1- Using the optomechanical components mounted on the optical breadboard, make an easy optical set-up to send the laser into the sphere.
2- Take the 3-step spectra as indicated previously and find the QE. See Figure below for an overview of the optical set-up to put up.

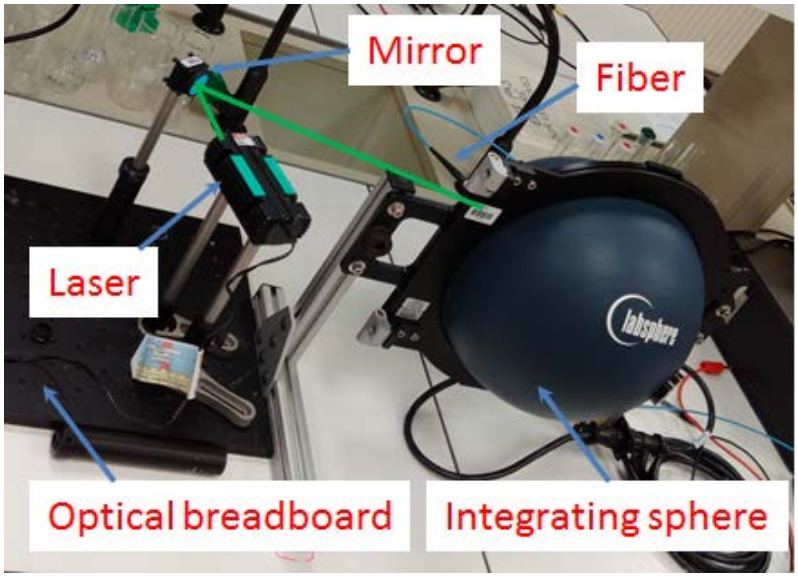

Figure 6: Picture of the quantum efficiency set-up. The rhodamine is inside the integrating sphere.

3- Compare with the literature. What could be the reason for the discrepancy?

QE of a blue LED and if time a white LED

Below is given what the experiment would be. In this case, we look into the measurement of the quantum efficiency of a blue LED. In fact, this is not exactly the quantum efficiency that will be measured. The quantum efficiency for an electroluminescent device is the ration of the number of emitted photons divided by the number of injected electrons. While the number of photons can be retrieved via a normalized calibrated spectrum, the number of electrons to be retrieved from the current, required to know the speed of the electrons in the material (I=nevS, with n the number of electrons per unit of volume and S the cross section of the wire). Instead of measuring the absolute QE we will measure what is also called QE but corresponds to a conversion ratio in energy. More precisely, the QE is the ratio of the total power radiated by the LED, divided by the electrical power:

$$QE = \frac{\phi_T}{\wp_{LED}} = \frac{\phi_T}{I_{LED}.V_{LED}}$$

To remind you, a Light-Emitting-Diode is made of semiconductors and is a p-n junction that emits light under an applied electric field.

1- Remove the optical fiber attached to the integrating sphere and replace it by the photodiode provided. Install the LED inside the integrating sphere and connect it to the source-meter that deliver I_{LED}. What is the relationship between Φ_T and the voltage detected by the photodiode?
2- Estimate the inner surface of the sphere and find the total optical power radiated by the LED.
3- Read I_{LED} and V_{LED} on the source-meter and give a value for the QE.
4- Does the measured QE agree with the literature?